SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS (SCORE)

WINTER SEMESTER 2025-26

Information Security Management CSE3502

REVIEW - 1

TITLE: IOT-BASED SMART ACCESS CONTROL & SECURITY MONITORING SYSTEM

Faculty: Dr. Priya V

TEAM DETAILS:

Yuvashree.G - 22MIS0300

Sachin.V-22MIS0303

2. Abstract

This project introduces a smart and secure access control system that combines RFID technology with face recognition and security monitoring features. Traditional access systems that depend only on RFID cards can be exposed to unauthorized use, card copying, and security issues. To address these problems, this project suggests a multi-factor authentication system that improves both safety and dependability. The system uses an ESP32 microcontroller to read RFID card data and sends it to a Flask-based backend server. The server confirms the user's identity by using a MySQL database and performs face recognition through the Local Binary Pattern Histogram (LBPH) machine learning algorithm. Additionally, the system applies role-based and time-based access control to make sure that users can only enter the system during allowed time slots and according to their assigned roles.

A key feature of this project is the Denial of Service (DoS) protection, which employs rate-limiting techniques. The system identifies repeated rapid access attempts and temporarily blocks suspicious users while generating alerts displayed on a web-based dashboard. The proposed system was tested in real-time situations and showed better security than traditional methods. It offers an efficient, cost-effective, and scalable solution for modern access control systems and has potential for future improvements using advanced deep learning models and cloud-based monitoring.

3. Keywords

RFID, Face Recognition, LBPH, IoT Security, Access Control, DoS Protection, ESP32

4. Introduction

Background of the Problem

In modern institutions and secure environments, access control systems are crucial for protecting resources, data, and infrastructure. Most traditional systems use RFID technology, where users gain access by scanning a card. While this method is convenient, it has major security issues. RFID cards can be lost, stolen, duplicated, or shared with unauthorized users, which makes the system vulnerable to abuse. These systems only verify card possession, not the actual identity of the user, so they do not provide complete security. Additionally, traditional systems do not have ways to detect unusual behaviors, such as repeated access attempts or misuse of valid credentials. These shortcomings show the need for a smarter and more reliable access control system that combines

identity verification and security monitoring to ensure that only authorized individuals can access resources under the right conditions.

Importance of the Study

This study is important because it aims to improve the security and reliability of access control systems by introducing multi-factor authentication and smart monitoring. In sensitive areas such as laboratories, research centers, and schools, unauthorized access can lead to serious risks, including data breaches and equipment misuse. By combining RFID-based identification with facial recognition, the system ensures both possession and identity verification, which greatly reduces the chances of unauthorized entry. Furthermore, adding time-based and role-based access control lets administrators limit access based on specific conditions, which improves overall management. Implementing Denial of Service (DoS) protection adds another security layer by detecting and stopping unusual or malicious access attempts. This study shows how combining IoT, machine learning, and cybersecurity techniques can create a more secure and efficient access control system suitable for real-world use.

Scope of the Project

This project focuses on designing and implementing a smart access control system using IoT and machine learning technologies in a controlled setting. The system combines RFID-based identification with facial recognition using the LBPH algorithm to provide multi-factor authentication. It also includes role-based and time-based access control to manage user entry. The project uses an ESP32 microcontroller for data collection, a Flask-based server for processing requests, and a MySQL database for storing user and access information. Additionally, a web-based dashboard is developed to monitor system activity and show security alerts. Basic DoS protection is also included using rate-limiting techniques to prevent repeated unauthorized access attempts. However, the system is limited to local deployment and does not feature cloud integration, mobile applications, or deep learning-based recognition models.

Objectives

The main goal of this project is to create a secure and smart access control system that addresses the limitations of traditional RFID-based systems. The system seeks to combine RFID technology

with facial recognition for multi-factor authentication, ensuring both identity and authorization. Another key goal is to implement role-based and time-based access control, allowing users to access the system only under allowed conditions. The project also aims to boost system security by detecting and blocking DoS attacks through rate-limiting methods. Additionally, it plans to offer a user-friendly web dashboard that provides access logs and security alerts in real time for effective monitoring. The system is designed to be affordable, efficient, and scalable, making it suitable for use in schools and laboratories. Overall, the goal is to enhance security, reliability, and usability in access control systems with modern technologies..

5. Literature Review

S.No	Research Paper (Title, Author, Year)	Research Contribution	Key Techniques Used	Limitations Identified	Research Gap
1	**RFID Based Smart Door Access Control System**, Kumar et al., 2021	Developed an RFID-based automated door security system for institutions.	RFID tags, Microcontroller, Relay Lock	No biometric verification, vulnerable to card sharing.	Need multi-factor authentication with identity verification.
2	**Face Recognition Based Attendance and Access System**, Sharma et al., 2020	Implemented face recognition for attendance and entry control.	OpenCV, LBPH, Webcam	Performance affected by lighting and pose changes.	Need integration with RFID for stronger authentication.
3	**IoT Enabled Smart Security System Using ESP32**, Ramesh et al., 2022	Introduced IoT-based real-time monitoring system with remote dashboard.	ESP32, WiFi, Flask, Sensors	No access control intelligence or biometric security.	Need secure access management with monitoring.

S.No	Research Paper (Title, Author, Year)	Research Contribution	Key Techniques Used	Limitations Identified	Research Gap
4	**Multi-Factor Authentication Using RFID and Biometrics**, Patel et al., 2019	Combined RFID with biometric authentication for secure entry.	RFID, Fingerprint Sensor	Costly hardware, fingerprint wear issues.	Face recognition can provide contactless authentication.
5	**LBPH Face Recognition for Real-Time Applications**, Ahonen et al., 2006	Proposed LBPH model for efficient face recognition.	Local Binary Pattern Histogram	Lower accuracy compared to deep learning in complex environments.	Can be improved with IoT integration and access control logic.
6	**Role-Based Access Control in Smart Campuses**, Singh et al., 2021	Designed user-role based permission system for campus resources.	Role-Based Access Control (RBAC), Database Rules	No hardware authentication layer.	Need RFID + face recognition with RBAC.
7	**Time-Based Authentication System for Restricted Areas**, Mehta et al., 2020	Restricted user access based on working shifts and schedules.	Time Scheduling, Database Validation	Only software-based checks, no identity verification.	Need real-time physical access system integration.
8	**Detection of DoS Attacks in IoT Networks**, Verma et al., 2022	Proposed mechanisms to detect flooding attacks in IoT systems.	Rate Limiting, Traffic Monitoring	Focused on network layer only.	Need DoS protection in access control systems.

S.No	Research Paper (Title, Author, Year)	Research Contribution	Key Techniques Used	Limitations Identified	Research Gap
9	**Web Dashboard for Security Monitoring Systems**, Joseph et al., 2021	Developed dashboard for logs, alerts, and live monitoring.	Flask, HTML, CSS, MySQL	No authentication or smart decision logic.	Need dashboard integrated with access control system.
10	**Smart Access Control Using RFID and Face Recognition**, Proposed Work, 2026	Combines RFID, LBPH face recognition, time control, role access, DoS protection, and dashboard monitoring.	ESP32, RFID, Flask, MySQL, LBPH, Rate Limiting	Prototype stage, local deployment only.	Future cloud deployment and deep learning enhancements.

6. Problem Statement

Description of the Problem

In many institutions and secure environments, access control systems are primarily based on RFID technology, where users gain entry using authorized cards. Although these systems are widely adopted due to their simplicity and cost-effectiveness, they suffer from significant security vulnerabilities. RFID cards can be easily lost, duplicated, or shared among unauthorized individuals, making it difficult to ensure that only the intended user is accessing the system. Furthermore, these systems do not verify the identity of the person carrying the card, leading to potential misuse. Another critical issue is the lack of monitoring mechanisms to detect suspicious activities such as repeated rapid access attempts, which may indicate a security attack. Without proper safeguards, these systems are prone to unauthorized access and misuse, highlighting the need for a more secure and intelligent solution that integrates identity verification and threat detection.

Target Users Affected

The limitations of traditional access control systems affect a wide range of users, particularly in environments that require controlled and secure access. Students in educational institutions may misuse shared or borrowed RFID cards to gain unauthorized entry into restricted areas such as laboratories. Faculty members and researchers working with sensitive equipment or confidential data are also at risk, as unauthorized access can lead to data breaches or equipment damage. Administrative staff responsible for managing access control systems may face difficulties in monitoring and identifying security threats due to the absence of real-time alerts and tracking mechanisms. Additionally, organizations that rely on basic access control systems are vulnerable to internal and external security threats. These issues impact overall safety, accountability, and operational efficiency, emphasizing the need for a more reliable and secure access control system.

Limitations of Existing Solutions

Existing access control solutions, particularly those based solely on RFID, have several inherent limitations. They rely on single-factor authentication, which only verifies possession of a card and not the identity of the user, making them highly susceptible to misuse. Biometric systems, although more secure, often involve higher costs and complex implementation, limiting their adoption in small-scale environments. Many IoT-based systems lack advanced security features such as anomaly detection and DoS attack prevention, leaving them vulnerable to cyber threats. Additionally, most systems do not incorporate real-time monitoring or alert mechanisms, making it difficult to detect and respond to suspicious activities promptly. Environmental factors such as lighting conditions can also affect the accuracy of biometric systems like face recognition. These limitations highlight the need for a cost-effective, integrated solution that combines multiple authentication methods with enhanced security features.

7. System Overview / Proposed Approach

System Description

The proposed system is a smart access control solution that combines RFID technology with face recognition and security monitoring to provide multi-factor authentication. The system uses an ESP32 microcontroller to read RFID card data and send it to a Flask-based server via WiFi. The

server verifies user details from a MySQL database and performs face recognition using the LBPH algorithm. It also checks role-based and time-based access permissions before granting entry. Additionally, the system includes DoS protection with rate-limiting techniques to detect repeated access attempts. A web-based dashboard displays access logs and security alerts in real time.

Objectives of the Proposed System

The main goal of the proposed system is to improve access control security by integrating multiple authentication methods, including RFID and face recognition. It aims to ensure that only authorized users can access the system by verifying both identity and credentials. Another goal is to implement role-based and time-based restrictions to control access based on user permissions and schedules. The system also focuses on enhancing security by detecting and preventing DoS attacks through monitoring repeated access attempts. Additionally, it aims to provide a user-friendly interface for monitoring system activity and alerts, ensuring efficient management and greater overall system reliability

8. Methodology

Data Collection (Survey/Dataset)

The system uses a custom dataset gathered from authorized users for face recognition. For each user, about 50 to 100 facial images are captured under various lighting conditions and angles using a webcam. These images are organized in labeled folders for each user.

The dataset trains the LBPH face recognition model. Additionally, user details such as name, UID (RFID), role, batch, and email are stored in a MySQL database. No external survey is conducted because the system relies on real-time data collection and user enrollment. This method ensures that the model is designed for the specific users of the system.

Tools/Technologies Used

The project uses a mix of hardware and software tools to implement the system. These technologies enable real-time processing, communication, and security.

Category	Tools/Technologies Used
Hardware	ESP32, RFID RC522, Camera
Backend	Flask (Python) Flask (Python)
Database	MySQL
ML/Computer Vision	OpenCV, LBPH
Frontend	HTML, CSS
Communication	HTTP (WiFi-based)

These tools are chosen for the efficiency, low cost, and ease of integration in IoT-based applications.

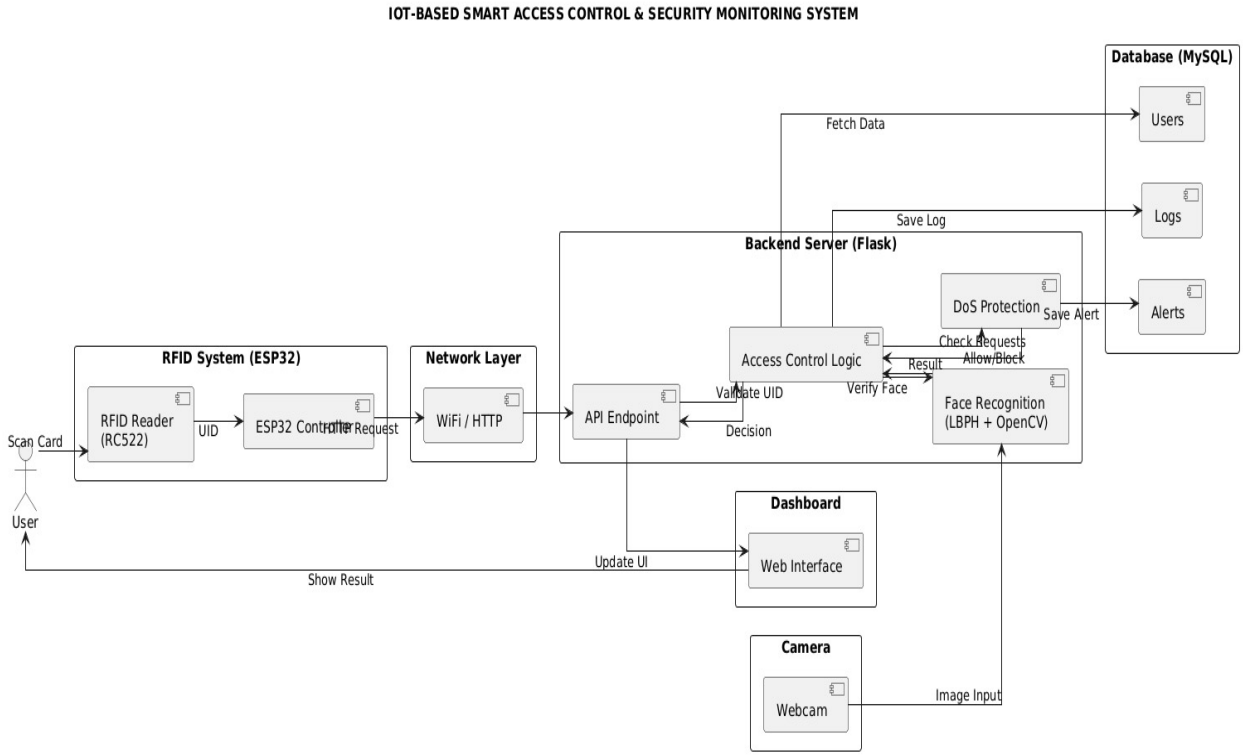
Step-by-Step Workflow

The system follows a clear workflow to ensure secure access control. The process starts with RFID scanning and concludes with access decision and logging.

Workflow Steps:

1. User scans RFID card using ESP32
2. ESP32 sends UID to Flask server via HTTP
3. Server checks user details in MySQL database
4. Camera captures user face
5. LBPH model performs face recognition
6. System checks role-based and time-based access
7. DoS protection checks for repeated attempts
8. Access is granted or denied
9. Logs and alerts are stored and displayed This workflow ensures a secure and efficient authentication process.

9. System Architecture



User

The user interacts with the system by scanning an RFID card to request access. The system verifies the user's identity through several authentication steps before granting or denying access.

Input

The input layer includes the RFID reader and camera. The RFID reader captures the UID, and the camera captures the user's face for identity verification.

Processing

The processing layer operates on the Flask server. Here, user data is validated, face recognition is performed using the LBPH model, and access decisions are made based on set rules.

Output

The output appears on the web dashboard and system responses. It indicates whether access is granted or denied and logs all activities for monitoring.

Security Layer

The security layer features DoS protection and access control measures. It detects unusual access attempts, enforces time and role limits, and maintains system integrity.

10. Implementation / Design

Modules/Components

The system is divided into multiple modules, each responsible for a specific function in the access control process. These modules work together to ensure secure and efficient operation.

Module	**Description**
RFID Module	Reads UID from RFID card using ESP32
Communication Module	Sends UID to server via HTTP
Server Module	Processes request using Flask
Database Module	Stores user details and alerts (MySQL)
Face Recognition Module	Verifies identity using LBPH
Access Control Module	Checks role and time permissions
Security Module	Detects DoS attacks
Dashboard Module	Displays logs and alerts

System Working

The system operates in real time by integrating hardware and software components. When a user scans an RFID card, the ESP32 reads the UID and sends it to the Flask server. The server retrieves user details from the database and activates the camera for face recognition. The LBPH model compares the captured face with stored data to verify identity. Once verified, the system checks role-based and time-based access conditions. Additionally, it monitors repeated access attempts to detect potential DoS attacks. Based on all checks, the system either grants or denies access. The result is logged and displayed on the dashboard.

Flow/Logic

The system follows a logical sequence of steps to ensure secure decision-making. Each stage validates a specific condition before proceeding to the next.

1. Start system
2. Scan RFID card
3. Send UID to server
4. Verify UID in database
5. Capture face image
6. Perform face recognition (LBPH)
7. Match face with user
8. Check role and time conditions
9. Apply DoS protection (rate limiting)
10. Grant or deny access
11. Store logs and alerts
12. End process

This structured logic ensures accuracy, security, and reliability in access control.

11. Results & Analysis

Survey Results / System Outputs

The team tested the system in different scenarios to check its performance and reliability. They executed various test cases, including valid access, invalid face recognition, access outside permitted time, and rapid repeated scans. The results showed that the system granted access only when all authentication conditions were met. Unauthorized attempts were denied, and any suspicious activities triggered alerts. The DoS protection mechanism detected multiple rapid scans and temporarily blocked access. The system also logged all events in the database and displayed them on the dashboard. Overall, the outputs confirm that the system works well in real-time conditions.

Sample Results Table:

Test Case	Expected Result	Actual Result
Valid RFID + Face	Granted	Granted
Wrong Face	Denied	Denied
Outside Time	Denied	Denied
Rapid Scan (DoS)	Blocked	Blocked

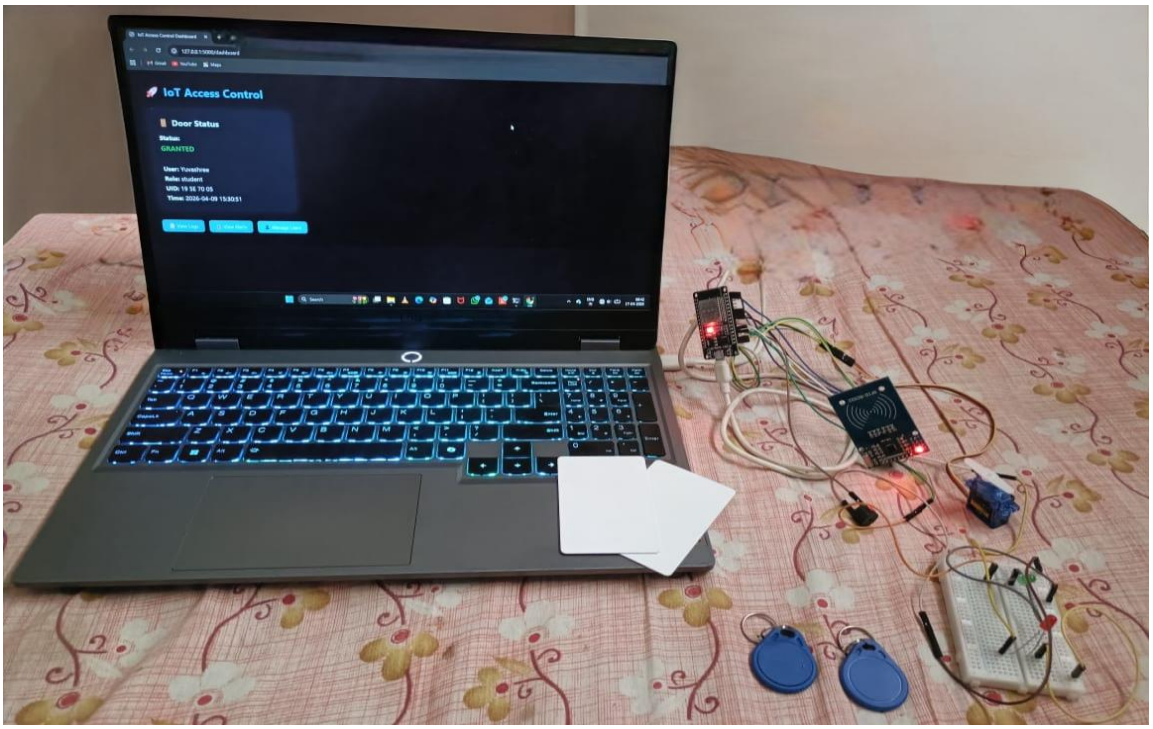
Charts/Graphs

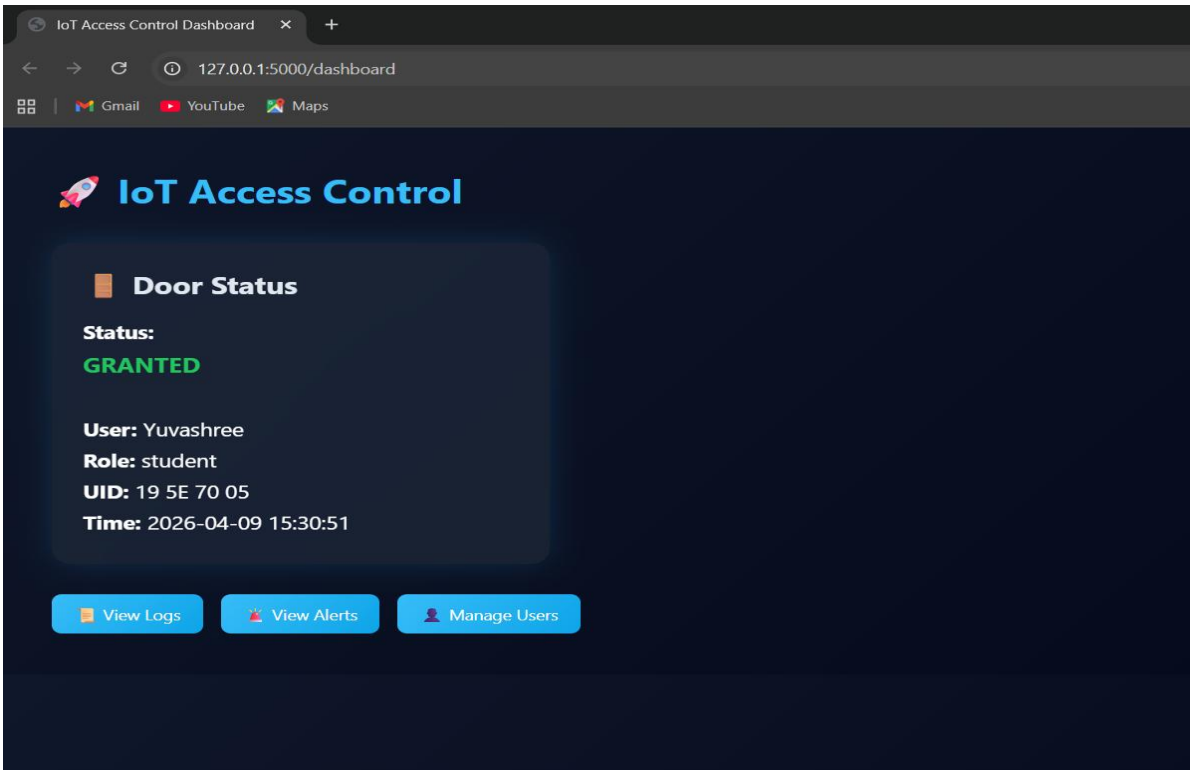
To better assess system performance, graphical representations can be used. A bar chart can compare different access outcomes, like granted, denied, and blocked requests. A pie chart can show the percentage of successful and failed authentication attempts. Additionally, a line graph can illustrate the frequency of access attempts over time, highlighting unusual spikes that suggest DoS attacks. These visuals aid in understanding system behavior and effectiveness. Tools like Excel or online graph generators can create these charts for inclusion in the report.

Key Observations

The results from the implementation reveal several important insights. The system improves security significantly compared to traditional RFID-only systems by adding face recognition as an extra authentication factor. The LBPH model works well even with a small dataset and offers real-time recognition. Time-based and role-based access control effectively limit unauthorized usage. The DoS protection mechanism can identify and stop repeated access attempts, further bolstering system security. However, the accuracy of face recognition may vary based on lighting conditions and image quality. Overall, the system presents a reliable and practical method for secure access control in real-world settings.

Screenshots:





Access Logs

Name	UID	Status	Time
Yuvashree	19 5E 70 05	GRANTED	2026-04-09 15:30:51
Yuvashree	19 5E 70 05	DENIED	2026-04-09 15:30:38
Yuvashree	19 5E 70 05	DENIED	2026-04-09 15:30:29
Sachin	B4 1E 9E A9	DENIED	2026-04-09 15:29:59
Sachin	B4 1E 9E A9	DENIED	2026-04-09 15:00:55
Yuvashree	19 5E 70 05	GRANTED	2026-04-09 15:00:45
Yuvashree	19 5E 70 05	DENIED	2026-04-09 15:00:20
Sachin	B4 1E 9E A9	DENIED	2026-04-09 15:00:11
Sachin	B4 1E 9E A9	DENIED	2026-04-09 14:59:59
Yuvashree	19 5E 70 05	DENIED	2026-04-09 14:59:45
Yuvashree	19 5E 70 05	DENIED	2026-04-09 14:56:57
Sachin	B4 1E 9E A9	DENIED	2026-04-09 14:56:37
Yuvashree	19 5E 70 05	DENIED	2026-04-09 14:56:13

Security Alerts

Name	UID	Message	Time
Unknown	19 5E 70 05	RESEARCHER 'Yuvashree' (evening batch) tried to access outside allowed hours	2026-04-07 12:03:32
Unknown	41 0C 97 04	scientist 'Priya Mam' tried outside allowed time	2026-03-13 23:12:09
Unknown	42 0F C0 01	admin 'System Admin' tried outside allowed time	2026-03-13 23:11:56
Unknown	B4 1E 9E A9	student 'Sachin' tried outside allowed time	2026-03-13 23:11:40
Unknown	19 5E 70 05	STUDENT 'Yuvashree' (evening batch) tried to access outside allowed hours	2026-02-20 12:31:57
Unknown	08 3D 9C A3	Unauthorized access attempt	2026-02-20 12:05:25
Unknown	19 5E 70 05	STUDENT 'Yuvashree' (evening batch) tried to access outside allowed hours	2026-02-20 11:52:57
Unknown	19 5E 70 05	STUDENT 'Yuvashree' (evening batch) tried to access outside allowed hours	2026-02-20 11:52:45
Unknown	D3 FD 5B F6	Unauthorized access attempt	2026-02-20 11:52:31
Unknown	19 5E 70 05	STUDENT 'Yuvashree' (evening batch) tried to access outside allowed hours	2026-02-20 11:52:04
Unknown	D3 FD 5B F6	Unauthorized access attempt	2026-02-20 11:41:45
Unknown	19 5E 70 05	STUDENT 'Yuvashree' (evening batch) tried to access outside allowed hours	2026-02-20 11:41:31
Unknown	19 5E 70 05	STUDENT 'Yuvashree' (evening batch) tried to access outside allowed hours	2026-02-20 11:35:52

12. Discussion

Interpretation of Results

The results show that the proposed system effectively boosts access control security by integrating multiple authentication methods. Combining RFID and face recognition ensures that both identity and authorization are verified before access is granted. The system performs reliably in real-time, correctly distinguishing between valid and invalid access attempts.

Implementing time-based and role-based restrictions strengthens access management by enforcing usage limits based on set conditions. Additionally, the DoS protection mechanism effectively identifies abnormal access patterns and prevents misuse. Overall, the results show that the system is a substantial improvement over traditional single-factor authentication systems in terms of security and reliability.

Strengths of the Solution

The proposed system has several strengths that make it effective and practical. Multi-factor authentication greatly enhances security compared to traditional RFID-only systems. Integrating IoT technology allows for real-time communication and monitoring. The LBPH-based face recognition model is lightweight, fast, and suitable for systems with limited computing power. Role-based and time-based access control add flexibility and better management of user permissions. Also, the DoS protection improves system security by detecting and blocking suspicious activities. The system is cost-effective, simple to implement, and scalable, making it suitable for educational institutions and small secure environments.

Limitations

Despite its strengths, the system has some limitations. The accuracy of the LBPH face recognition model can be influenced by variations in lighting, facial expressions, and camera quality. The system depends on a locally stored dataset, which may restrict its scalability and adaptability for larger user groups. Additionally, the DoS protection mechanism relies on basic rate-limiting and may not catch more complex cyberattacks. The system is currently designed for local use and lacks cloud integration or remote access features. Also, using basic hardware components may limit performance in high-security or large-scale environments, indicating a need for further improvements.

13. Future Enhancements

Possible Improvements

The system can be enhanced by adding advanced technologies to increase accuracy, security, and usability. One major enhancement is integrating deep learning models like Convolutional Neural Networks (CNN) or FaceNet for more accurate face recognition in various conditions. The system

can also be expanded to include OTP-based or mobile authentication for an extra security layer. Implementing encrypted communication between the ESP32 and the server could further enhance data security. A mobile or web application interface could simplify monitoring and control. Additionally, real-time notifications, like email or SMS alerts, can improve responsiveness to security events.

Scalability Aspects

The system can scale to support more users and multiple access points in different locations. Integrating cloud-based databases and servers would allow handling larger data volumes and provide remote access and monitoring options. Centralized management systems can enable administrators to manage multiple devices from a single interface. Optimizing the database and improving network communication can also enhance performance under heavy use. The architecture could extend to support additional biometric methods like fingerprint or iris recognition. These scalability upgrades make the system suitable for large institutions and organizations.

14. Conclusion

Summary of Work

This project focuses on creating a smart and secure access control system by combining RFID technology with face recognition and security monitoring. The system uses an ESP32 microcontroller to read RFID data and communicate with a Flask-based server. It implements face recognition using the LBPH algorithm for user identity verification. Additional features, like role-based and time-based access control, ensure users only access the system under specific conditions. The implementation also includes DoS protection through rate-limiting techniques. Overall, the system merges IoT, machine learning, and security mechanisms to provide a reliable and efficient access control solution.

Final Outcome

The final system showcases a multi-factor authentication approach that enhances security compared to traditional methods. It accurately verifies users with both RFID and face recognition before granting access. The system effectively enforces access restrictions based on user roles and

time schedules. The DoS protection mechanism identifies repeated access attempts and prevents misuse by temporarily blocking users and sending alerts. A web-based dashboard offers real-time monitoring of access logs and security events. The system operates reliably across various test scenarios, proving its effectiveness for use in controlled environments like laboratories and institutions.

Impact

The developed system significantly improves access control security by addressing the weaknesses of traditional RFID-based systems. By integrating multiple authentication methods, it ensures greater accuracy and reduces the chances of unauthorized access. The DoS protection enhances system resilience against unusual and malicious activities. The system boosts safety, accountability, and operational efficiency where secure access is vital. Its cost-effective and scalable design makes it suitable for educational institutions and small organizations. Additionally, the project sets a solid foundation for future advancements in smart security systems using IoT and machine learning technologies.

15. References

[1] K. Finkenzeller, “RFID Handbook: Fundamentals and Applications,” Wiley, 2010.

[2] P. Viola and M. Jones, “Rapid Object Detection using a Boosted Cascade of Simple Features,” IEEE, 2001.

[3] T. Ahonen, A. Hadid, and M. Pietikäinen, “Face Recognition with Local Binary Patterns,” IEEE, 2006.

[4] OpenCV Documentation, “Face Recognition using LBPH,” https://opencv.org

[5] Flask Documentation, “Flask Web Framework,” https://flask.palletsprojects.com

[6] Espressif Systems, “ESP32 Technical Reference Manual,” 2020.

[7] S. Kumar et al., “IoT-Based Smart Access Control System,” IEEE, 2021.

[8] N. Dalal and B. Triggs, “Histograms of Oriented Gradients for Human Detection,” IEEE, 2005.

[9] MySQL Documentation, “MySQL Reference Manual,” https://dev.mysql.com

[10] A. Jain et al., “Biometric Systems: Security and Privacy,” IEEE, 2016.

16. Appendix

Survey Questions

The following sample questions can gauge user perspectives on access control systems. These questions help assess the need for better security mechanisms. Users may be asked whether they currently use RFID-based systems and if they have faced any unauthorized access issues. Questions may also focus on awareness of security threats, such as card misuse or DoS attacks. Additionally, users can be asked if they prefer multi-factor authentication methods like face recognition. These responses help justify the need for the proposed system and highlight its relevance in real-world situations.

Additional Data / Screenshots

This section includes supporting materials like system screenshots and outputs to validate the implementation. Images of RFID scanning through the ESP32, face recognition during authentication, and results showing access granted or denied can be included. Screenshots of the alerts dashboard, displaying DoS detection and database tables with user and alert information, should also be provided. Each image should be labeled with figure numbers and captions for clarity. These visual elements enhance the understanding of the system's functionality and provide evidence of successful implementation and testing.

Plagiarism Declaration

I confirm that this project report is original and free from plagiarism.